

RADIANT: Better rPPG estimation using signal embeddings and Transformer

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Abstract

Remote photoplethysmography can provide non-contact heart rate (HR) estimation by analyzing the skin color variations obtained from face videos. These variations are subtle, imperceptible to human eyes, and easily affected by noise. Existing deep learning-based rPPG estimators are incompetent due to three reasons. Firstly, they suppress the noise by utilizing information from the whole face even though different facial regions contain different noise characteristics. Secondly, local noise characteristics inherently affect the convolutional neural network (CNN) architectures. Lastly, the CNN sequential architectures fail to preserve long temporal dependencies. To address these issues, we propose RADIANT, that is, rPPG estimation using Signal Embeddings and Transformer. Our architecture utilizes a multi-head attention mechanism that facilitates feature subspace learning to extract the multiple correlations among the color variations corresponding to the periodic pulse. Also, its global information processing ability helps to suppress local noise characteristics. Furthermore, we propose novel signal embedding to enhance the rPPG feature representation and suppress noise. We have also improved the generalization of our architecture by adding a new training set. To this end, the effectiveness of synthetic temporal signals and data augmentations were explored. Experiments on extensively utilized rPPG datasets demonstrate that our architecture outperforms previous well-known architectures. Code: <https://github.com/Deep-Intelligence-Lab/RADIANT.git>

1. Introduction

The heart continuously pumps blood through the capillaries and induces periodic cardiovascular pulse throughout the body. The number of pulses induced in a minute is known as heart rate (HR). HR estimation is important to measure a person's health [21]. It is a primary indicator of heart-related problems and mental health states ranging from stress, depression, anxiety, and excitement [21].

The non-invasive HR estimation techniques comprise of

electrocardiogram (ECG), ballistocardiogram (BCG) and photoplethysmography (PPG). These techniques require contact with the skin surface and cause discomfort for continuous HR estimation [61]. Hence, they offer limited applicability for skin-damaged patients, patients suffering from severe skin infections, exercise monitoring, and neonates monitoring [21, 29]. In contrast, remote Photoplethysmography (rPPG) is a non-contact HR estimation method utilizing non-contact face videos for estimating cardiovascular pulse. Since, it avoids any contact between sensors and skin, it can be used in applications where contact-based PPG methods cannot be used such as, drowsy driver detection [53], deepfake detection [25, 54], face anti-spoofing [4, 5], micro-expression recognition [16], micro-expression spotting [20, 17], lie detection [59] and stress monitoring [49, 38]. In the ongoing SARS-CoV-2 pandemic rPPG can be used for automated HR estimation and provide support to the patients requiring urgent and critical telehealthcare. It motivates us to propose an accurate rPPG-based HR estimation method.

The rPPG method analyzes the variations in the blood flow volume in the carotid arteries beneath the facial skin [21]. These variations induce subtle skin color changes imperceptible to the human eyes [34]. However, a video camera is capable of capturing these variations. The face videos acquired in controlled environments provide relevant rPPG information, resulting in correct HR estimation. However, in real scenarios, these videos are influenced by noise due to facial movements, illumination variations [51], and other artifacts and thus, result in spurious HR estimation [22].

The rPPG-based HR estimation first requires the extraction of the relevant face region, known as the region of interest (ROI). Usually, the color variations are analyzed in the ROI and the corresponding signals are referred to as temporal signals. The temporal signals are obtained by either averaging pixel values in the face ROI [39], or subtracting the pixel values from the consecutive video frames [9]. Eventually, the cardiovascular pulse present in the temporal signals is extracted using domain-specific knowledge. To this end, [45] and [2] have employed blind source separation (BSS) and maximum periodicity criteria for HR estimation [45].

Unfortunately, their HR estimations are erroneous when the temporal signals contain periodic noise.

Apart from domain-specific knowledge, rPPG-based HR estimation can be performed using deep learning. For instance, [9, 43] have utilized Convolutional Neural Network (CNN) based architectures for rPPG estimation. Further, AND-rPPG [3] have used temporal convolution network for pulse estimation. Further, a combination of CNN and sequential architecture is used in [39] for rPPG estimation. Unfortunately, the performance of these architectures can be affected by slight facial movements persisting in small facial regions, even for a short duration. Such behaviour is attributed to the local feature encoding of the CNNs [31]. Moreover, the sequential architectures are incompetent to provide correct HR estimation as these architectures fail to model long temporal dependencies [13, 37]. These issues are alleviated by employing dual-stream Transformer architecture in [30]. However, the background color variations used for denoising, fail to provide effective noise representation [32]. Additionally, many features are required by [30] when temporal signals are extracted using the difference of frames. The proper training of the corresponding architectures necessitates large-scale datasets to alleviate the problem of underfitting [7]. Unfortunately, the datasets available for training have limited training data, which restricts the applicability of frame difference-based temporal signal extraction.

We propose a novel rPPG-based HR estimation method, *RADIANT*, that is, better rPPG estimation method using signal embeddings AND Transformer. We obtain the temporal signals by averaging the skin color variations in ROIs that provides efficient rPPG feature representation allowing convergence over limited training data. Furthermore, we utilize the Transformer architecture for estimating pulse as it can learn global context and effectively mitigate local noise [56]. Our main contributions are:

(1) Our proposed rPPG architecture utilizes Multilayer perceptron (MLP) for projecting the temporal signals into signal embeddings and the attention processing ability of the Transformer architecture. Linear projection using MLP provides the proper learning of relevant feature representation for rPPG information while the Transformer architecture effectively performs denoising and cardiovascular pulse estimation. (2) To mitigate the problem of limited training data, we explore the possibility of pre-training our Transformer architecture using synthetic temporal signals [39] and data augmentation [42]. Both are performed in a time-efficient manner, and it is observed that they improve the performance by allowing domain adaptation. (3) Our experimental results demonstrate that we obtain state-of-art results on publicly available datasets.

2. Related Works

2.1. Domain knowledge-based rPPG methods

The rPPG-based HR estimation can be performed by utilizing domain knowledge. Such estimation involves the following steps: ROI detection, spatial filtering, temporal signal extraction, and pulse signal estimation. Usually, color variations are employed to define the temporal signals. Amongst the RGB colors, the green color channel is shown to be the aptest for rPPG information extraction in [57]. On the contrary, a chrominance subspace transformation of the RGB color signals is used for pulse estimation in [10]. Further, BSS is used in [45] for pulse estimation. Eventually, HR is given by the peak frequency in the Fourier power spectrum of the pulse signal [3]. The above-discussed methods are incompetent to distinguish pulse signal and noise because they utilize handcrafted representations to model noise, and they lack the appropriate supervision to understand the noise attributes induced by facial movements [61].

2.2. CNN and sequential architectures for rPPG

CNN architectures have been used extensively in rPPG-based HR estimation because they allow feature subspace mapping with minimal requirement of domain-specific knowledge. For instance, [27] feeds the time-frequency representation of chrominance signals into the VGG15 [50] for HR estimation. A depthwise separable CNN architecture is used in [46] for pulse signal estimation from the Spatio-temporal feature representation of the color variations. Recurrent Neural Network (RNN) is employed by [40] for modeling time dependencies between the temporal signals to estimate HR. ETA-rPPGNet [28] utilizes the time domain subnet to address the problem of redundant rPPG information and noise induced by slight facial deformations. Moreover, the rPPG methods in [9, 35] have utilized end-to-end architectures for extracting relevant rPPG information from face videos. The method in [9] feeds the normalized frame difference into a CNN for HR estimation. Similarly, encoder-decoder architecture is employed in [41] for learning noise and rPPG information. Training these huge architectures is challenging, and it tends to underfitting over small-scale datasets [7]. Temporal difference CNN is used in [36] for capturing temporal color variations and generating appropriate signal representation. The importance of synthetic temporal signal generation in rPPG-based HR estimation has been explored in [39]. The synthetic signals can be efficiently generated for pre-training using periodic sine curves and random noise.

The CNN or sequential architectures require diverse datasets in considerable quantities to alleviate underfitting [7]. Further, they fail to learn global information from the facial regions and fail to model the long temporal dependency [18] if sequential architectures are employed [40].

2.3. Attention mechanism and Transformer

The Transformer architecture is proposed in [56] for Natural Language Translation. It models the complex global contextual dependencies between the words in a sentence using multi-head attention mechanism. The huge success of Transformer architecture due to the appropriate modeling enables their applicability in various natural language processing tasks [11, 14, 15]. The applicability of these architectures has been further explored for the image classification task using the Vision Transformer (ViT) [12], and performance improvements have been examined in this direction. It proliferates the research in unraveling the effectiveness of the Transformer architectures for computer vision applications [55, 8]. The rPPG-based HR estimation is no different. It aims to estimate pulse signals from several temporal signals, where each temporal signal contains mainly the pulse signal corrupted by noise. Thus, the pulse signal results in a strong correlation between the temporal signals, which the Transformer architecture can easily learn. For instance, TransPPG proposed in [30] feeds the temporal signals extracted from frame differences to the Transformer architecture for estimating the pulse signal. Similarly, [47] have utilized the Transformer architecture [56] for estimating pulse signal from facial video.

Kindly note that these estimated pulse signals can be erroneous when the corresponding temporal signals contain significant noise. Furthermore, these methods require large parameter learning using limited training data for relevant pulse signal estimation. This situation results in an under-fitting problem and thereby degrade the efficacy [7].

3. Proposed method

This section presents our proposed rPPG-based HR estimation method, *RADIANT*. Figure 1 shows the flow diagram of our proposed method. Initially, the face region is identified and divided into multiple ROIs. Subsequently, the temporal signals are extracted from these ROIs, and chrominance subspace transformation [10] is applied to alleviate the effects of motion and luminance. Subsequently, an MLP layer is applied to project the resultant temporal signals into signal embeddings. Eventually, the Transformer architecture utilizes the signal embeddings for pulse estimation.

3.1. Video Clip Extraction

For HR estimation, the video is divided into multiple clips. We have utilized a non-overlapping window over the video clips with a window of 4 seconds for HR estimation. Such division overcomes loss of information due to lack of complete pulse signal in a small time interval [28]. The mean of the HRs estimated for 5 consecutive video clips is obtained for obtaining the HR for a 20 second video.

3.2. Temporal Signal Acquisition

3.2.1 ROI Extraction

Our first step is to define a fixed ROI that provides relevant rPPG information from an input video clip. The relevant rPPG information is present in the facial skin regions. Thus, we employ facial landmark points that outline the boundaries of the face, including its subparts like eyes, lips, and nose. These landmark points are extracted using the CLNF Openface 2.2.0 landmark detector [62] for the first video frame because computing landmark points for each video frame is time expensive [18]. It provides 68 landmark points. Since the regions near the eyes tend to be easily influenced by facial expressions, we avoid them for defining the ROI [3]. Similarly, the forehead region is avoided as it is usually covered with hairs to induce noise in the corresponding temporal signals [61]. We mainly use the region below the eyes containing the cheeks for obtaining the temporal signals. The desired face region containing relevant rPPG information is obtained by computing the convex hull of the landmark points: 1) 2, 3, 4, 5, 6 of the left cheek, 2) 12, 13, 14, 15, 16 of the right cheek, 3) 29 of the nose, and 4) 7, 8, 9, 10, 11 of the chin. It is observed in [3] that the face boundaries are prone to significant temporal variations from minimal facial deformations. Thus, we employ the morphological operation to remove the boundary pixels similar to [62]. For a better visualization, kindly refer to the supplementary material.

We can extract the temporal signal from the entire face region, but noise-induced in a small facial region affects the resultant temporal signal. Thus, it is recommended in [18] to divide the region into smaller ROIs, extracting temporal signals from those ROIs, and then consolidate the signals for pulse estimation. Following this similar path, we divide the obtained facial region into small non-overlapping square blocks of the same size. Amongst the square blocks, we consider only those square blocks as ROI whose all the pixels belong to the skin pixels. The skin pixels are detected using the method described in [44]. We have used the method described in [22] for choosing the optimum size of our square blocks while alleviating the effect of scale difference among different faces.

3.2.2 Temporal Signal Extraction

The temporal signal extraction is performed in two steps: RGB signal extraction and projection into chrominance signals. The RGB signals are obtained by averaging the pixel values of the red, green, and blue channels from the face ROI. Mathematically, the temporal signal r_j , denoting the red color channel signal for j^{th} ROI is given by:

$$r_j = \left(\frac{\sum_k r_{j,1}^k}{p_{j,1}}, \frac{\sum_k r_{j,2}^k}{p_{j,2}}, \dots, \frac{\sum_k r_{j,f}^k}{p_{j,f}} \right) \quad (1)$$

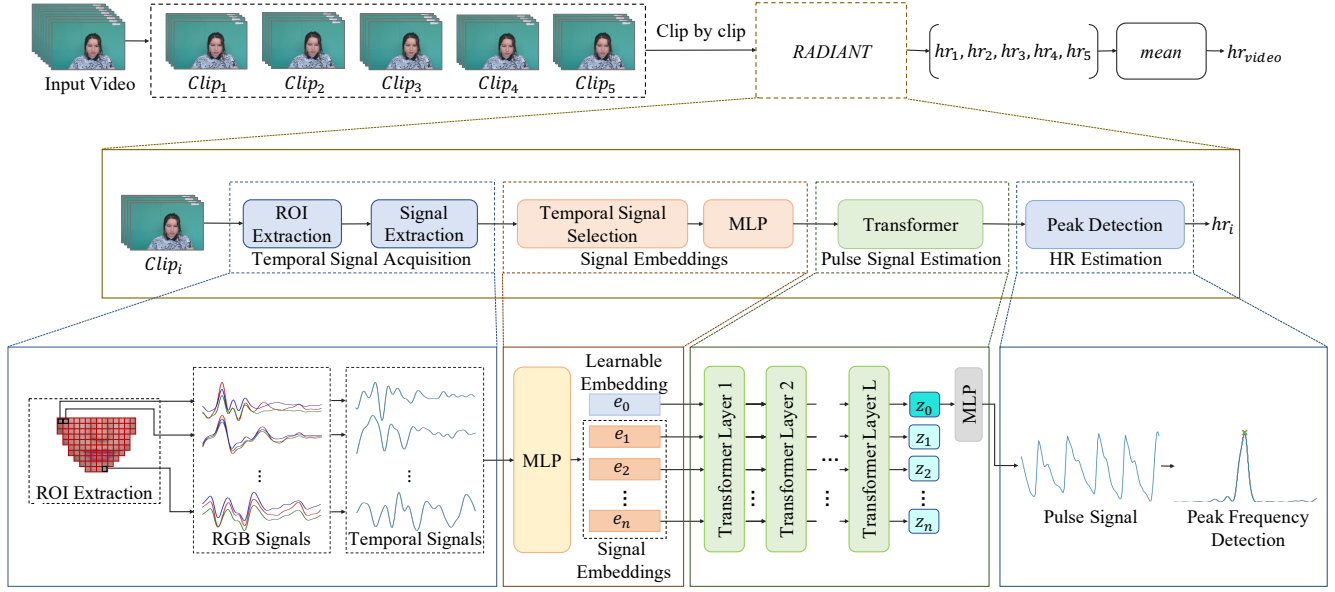


Figure 1. The flow diagram of our proposed method *RADIANT*.

where, $r_{j,i}^k$ refers to the red color channel intensity of the k^{th} pixel in the j^{th} ROI of the i^{th} frame. Similarly, the temporal signals g_j and b_j are obtained from the green and blue color channel intensities respectively. The total number of pixels in j^{th} ROI of i^{th} frame is represented by $p_{j,i}$ and the total number of frames in the video clip is represented by f . The obtained RGB signals are then passed through a bandpass filter to obtain the filtered signals as $\tilde{r}_j$, $\tilde{b}_j$ and $\tilde{g}_j$. Mathematically:

$$\tilde{r}_j = \psi_{bp}(r_j), \tilde{g}_j = \psi_{bp}(g_j), \tilde{b}_j = \psi_{bp}(b_j) \quad (2)$$

where, $\psi_{bp}(\cdot)$ is a Butterworth bandpass filter of order 4 [60] that suppresses any signal components that correspond to frequencies outside the HR range (0.7 Hz to 4.2 Hz) [3].

These RGB signals are then projected into chrominance signals to minimize the noise artifacts and suppress specular reflections [10]¹. More details on chrominance transformation are provided in the supplementary material. The resultant temporal signal is filtered using a detrending filter to mitigate noise due to illumination variations [21].

3.3. Signal embeddings

3.3.1 Temporal signal selection

Assume that $(c_1, c_2, \dots, c_m)$ are the temporal signals obtained from m facial ROIs. Kindly note that different video clip results in a different number of ROIs. That is, the value of m depends on the input clip. Usually, deep learning architectures require fixed-size input. Thus, we select

n temporal signals amongst the m extracted temporal signals. The value of n is selected such that each video clip contains at least n ROIs. For selecting the temporal signals, we use the observation that some extracted temporal signals are affected by noise due to facial movements. For instance, motion caused due to smiling usually affects the temporal signals obtained from the regions near the lips while leaving the other ROIs unaffected. Thus, we choose those n temporal signals that are least affected by facial deformations. To this end, we leverage the intuition that the face regions with facial movements will have large skin color variations, resulting in large standard deviation in the amplitude of the extracted temporal signals. Thus, temporal signals with less standard deviation will provide better rPPG information. Hence, we select those top- n temporal signals which have the higher quality scores, where the quality score corresponding to temporal signal c_j is given by:

$$quality_j = \frac{1}{\sigma(c_j)} \quad (3)$$

where σ refer to the standard deviation operator.

3.3.2 MLP

Even though the chosen temporal signals are least affected by noise, they can still result in erroneous HR estimation because they contain noise. Thus, we project the temporal signals into signal embeddings $(e_1, e_2, \dots, e_n)$ of higher dimensions using an MLP. This projection allows adequate representation subspace for learning relevant rPPG features and performing denoising. Furthermore, we will utilize the

¹Code <https://github.com/phuselab/pyVHR>

Transformer architecture for pulse estimation. The architecture consists of multiple Transformer layers that require input vectors of the same dimensions d . We want to benefit from effective weight initialization by pre-training with ImageNet [48] dataset. Hence, we have set the dimensions to 768 as the pre-trained weights require the input of size 768. We utilize a learnable MLP layer with 768 output nodes for projecting a temporal signal c_j into an embedding e_j , for $j = 1, 2, \dots, n$. Kindly note that our projection ensures that rPPG information in an embedding only depends on the corresponding temporal signal while remaining unaffected by other temporal signals. The reason for choosing such a projection is that when an embedding depends on several temporal signals, then information from a temporal signal with higher noise can easily affect the embedding, leading to the incorrect HR estimation [3]. Finally, inspired by the use of classification token by the architectures in [12, 11], we have prepended a learnable embedding $e_0 \in \mathbb{R}^{1 \times d}$ that learns the feature representation of the pulse signal.

3.4. Pulse signal estimation

3.4.1 Transformer Architecture

Each signal embedding contains a composition of features from pulse signal, other physiological parameters, and noise. Thus, we need to consolidate and filter the rPPG information from the signal embeddings for correct pulse signal estimation. To this end, we employ the Transformer architecture. The Transformer architecture utilizes the self-attention mechanism for consolidating feature information from the input components [31]. The self-attention mechanism learns the contextual dependencies between the signal embeddings, allowing our architecture to learn the correlations for contributing to rPPG information. Subsequently, the consolidated rPPG features are transformed using a two-layer MLP [31]. This combination allows the Transformer architecture to denoise and consolidates rPPG features for correct pulse signal estimation.

The self-attention mechanism is the building block of the Transformer. For each signal embedding e_j , it computes a corresponding latent vector z_j which is the weighted sum of all the signal embeddings. Here, z_j is of the same dimensions as e_j . For obtaining these corresponding weights, the signal embeddings e_j are projected into query ($q_j \in \mathbb{R}^{1 \times d_q}$), key ($k_j \in \mathbb{R}^{1 \times d_k}$) and value ($v_j \in \mathbb{R}^{1 \times d_v}$) vectors using learnable weights. Alternatively, the signal embedding matrix $E \in \mathbb{R}^{n \times d}$ containing all the signal embeddings is projected into matrices: $Q \in \mathbb{R}^{n \times d_q}$, $K \in \mathbb{R}^{n \times d_k}$ and $V \in \mathbb{R}^{n \times d_v}$ corresponding to the queries, keys and values, respectively. That is,

$$Q = E \cdot W^Q, K = E \cdot W^K, \text{ and } V = E \cdot W^V \quad (4)$$

$$W^Q \in \mathbb{R}^{d \times d_q}, W^K \in \mathbb{R}^{d \times d_k}, W^V \in \mathbb{R}^{d \times d_v}$$

where, W^Q , W^K and W^V are the learnable weights.

The correlations between the signal embeddings are reflected by the attention scores obtained by:

$$SA = \text{softmax} \left(\frac{Q \cdot K^T}{\sqrt{d_q}} \right) V \quad (5)$$

where, the dot product between its query and all the keys is calculated for a given signal embedding. The resulting value is scaled by the square root of d_q followed by a softmax operation. The obtained scores associated with each embedding transform them into a weighted sum of the features from all the signal embeddings.

There are multiple relationships among the signal embeddings contributing toward rPPG information. For encapsulating such relationships, the multi-headed self-attention (MSA) mechanism is employed by the Transformer architectures, as in natural language processing [11] and computer vision [12] applications. The MSA mechanism transforms the queries, keys, and values into multiple learned linear projections, corresponding to multiple heads, h , for modeling multiple relationships. Subsequently, independent self-attention computation is performed in the respective heads using the equation 5 resulting in $SA_0, SA_1, \dots, SA_{h-1}$. Kindly note that, here SA_i for $i = 0, 1, \dots, h-1$ represents the output from i^{th} head. Finally, for consolidating the information, the output obtained from each head is concatenated and projected into d dimensional vectors using a learnable matrix W^O . That is, the matrix containing the latent vectors, Z' is given by:

$$Z' = [SA_0, SA_1, \dots, SA_{h-1}] \cdot W^O, W^O \in \mathbb{R}^{(h \cdot d_v) \times d} \quad (6)$$

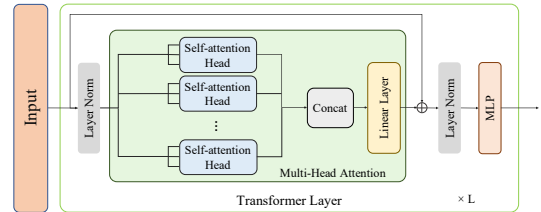


Figure 2. An illustration of a Transformer layer.

The output Z' is then fed to a LayerNorm (LN) layer [1] followed by a two layer MLP with GELU activation function [24] for feature transformations. A residual connection is placed after the MLP output. The architecture of a single Transformer layer is depicted in Figure 2. Kindly note that there are multiple Transformer layers and the output from a Transformer layer l is given by:

$$Z_l = MLP(LN(MSA(Z_{l-1}))) + Z_{l-1}, \text{ where} \quad (7)$$

$$Z_0 = [e_0, e_1, \dots, e_n]$$

Assume that the output of the last Transformer layer $\mathbf{Z}_L$ is $(z_0, z_1, \dots, z_n)$. Among these vectors, the vectors $(z_1, z_2, \dots, z_n)$ contain the refined correlations that contribute to the rPPG information. However, the vector z_0 contains the pulse information extracted from the correlations learned by the other latent vectors $(z_1, z_2, \dots, z_n)$. We use the mean square error loss function during training between the estimated pulse and the ground truth. It necessitates that the dimensions of the estimated and ground-truth pulse signal should be the same. Thus, an MLP head is attached to z_0 that transforms it into the pulse signal of the same dimensions as the ground truth.

3.4.2 Pre-training

Transformers show poor generalization capabilities in the computer vision domain due to the lack of their inductive bias [12]; thus, pre-training is employed to improve their generalization. Following this, we employ pre-training in two stages. The first stage performs the pre-training over ImageNet dataset [48]. While the second stage uses synthetic temporal signals generated using the method described in [39]. For brevity, the synthetic signals are obtained using sine waves. Since the waves should represent periodic cardiovascular pulses, their frequencies are set between 0.7 Hz to 4.2 Hz, corresponding to the normal HR range. In addition, the generated sine waves are superimposed with another sine wave simulating breathing rhythm with periodicity between 5 beats per minute (BPM) to 20 BPM, random step signals and Gaussian noise. It stimulates the noisy variations. Further details are provided in supplementary material. After pre-training on the synthetic dataset, we perform fine-tuning on public rPPG datasets.

3.4.3 Data Augmentation

In the rPPG datasets, most samples contain HR ranges from 60 to 90 BPM, resulting in uneven data distribution. Training on such dataset will make our architecture biased towards samples containing HR in this range. Addressing this issue [42] have described a novel data augmentation method comprising temporal upsampling and downsampling of the video for providing proper HR coverage in the training dataset. Alternatively, temporal signal interpolation can be employed to obtain the same effect, and such interpolation prevents the time-consuming steps of processing video frames. Thus, we have utilized this technique to perform our data augmentation. Specifically, we generate samples with higher HR ranges by downsampling the temporal signals by 2 and 3 times. Similarly, we generate the samples with lower HR values by upsampling the temporal signals twice and thrice. Also, we have discarded the augmented samples, whose HR values lie outside the human HR range of 40 – 240 BPM [3].

3.5. Heart Rate Estimation

The raw pulse signal ($\mathbf{y}$) obtained from the last Transformer layer is bandpass filtered for removing any signal components whose frequencies lie outside the normal HR range. We have applied the bandpass filter $\psi_{bp}(\cdot)$ (described in 3.2.2) over $\mathbf{y}$ to obtain the clean pulse signal $\tilde{\mathbf{y}}$. Subsequently, the pulse spectrum ($PS[freq]$) is obtained by applying Fast Fourier transform (FFT) over the pulse signal $\tilde{\mathbf{y}}$. Note that $PS[freq]$ is the amplitude of the pulse spectrum at frequency $freq$. For a video clip i , the HR is given by:

$$hr_i = \underset{freq}{\operatorname{argmax}} PS[freq] \times 60 \quad (8)$$

kindly note that as stated in section 3.1, the HR for a 20 seconds video is obtained by averaging the HR obtained from the consecutive 5 short video clips. Hence, the final HR estimate for a 20 seconds video is given by:

$$hr_{video} = \operatorname{mean}(hr_1, hr_2, hr_3, hr_4, hr_5) \quad (9)$$

4. Experimental Results

4.1. Dataset and Metrics

We have provided evaluation results for the public UBFC-rPPG [6] and COHFACE [26] datasets. The UBFC-rPPG dataset consists of facial videos of 2 minutes from 42 subjects. Videos are recorded in a resolution of 640×480 in 8-bit RGB format at the frame rate of 30 frames per second. We have split the dataset into training and testing sets with videos from 28 subjects into training set and 14 subjects into the testing set. The COHFACE dataset contains face videos of 40 subjects alongwith their physiological information. Each video is of 1 min recorded at 20 frames per second. For reporting our evaluation results, we have reported mean absolute error (MAE), standard deviation (σ) and root mean squared error (RMSE) between the ground truth HR and estimated HR.

4.2. Choice of Training Parameters

We have used Adam optimizer with a learning rate of 3×10^{-4} . We have used a batch size of 32 for pre-training our architecture using the synthetic temporal signals. However, for fine-tuning, we have used a batch size of 4. We have performed pre-training and fine-tuning over 20 and 50 epochs, respectively. We have used the mean squared error loss function for training our architecture.

4.3. Comparative evaluation

We have provided the comparison of our method, *RADIANT*, with previous rPPG methods on UBFC-rPPG and COHFACE datasets. Note that we have used publicly available codes and experimental settings for comparisons. Furthermore, we have used the standard training and testing split

Table 1. Performance evaluation of *RADIANT* for Average HR variation per video. All the values are in BPM and all the metrics represent better performance if they have lower values.

	UBFC-rPPG			COHFACE		
	σ	MAE	RMSE	σ	MAE	RMSE
[58]	17.89	15.95	11.65	22.30	20.97	25.98
[45]	12.80	06.95	13.60	13.83	08.89	14.55
[10]	05.21	03.21	06.14	11.61	10.15	12.69
[18]	07.00	06.15	07.92	08.10	08.27	11.31
[19]	06.02	05.08	07.42	07.98	08.97	10.84
[33]	08.00	06.54	09.11	11.52	09.31	12.27
[9]	08.73	06.27	10.82	09.01	08.25	14.71
[52]	05.21	04.90	05.89	09.46	08.10	10.80
[47]	08.18	11.28	13.94	11.24	19.66	22.65
<i>Ours</i>	03.45	02.91	04.52	07.41	08.01	10.12

as described in [52, 26] for a fair comparison. Kindly note that the InstTrans [47] architecture provides HR estimation heart rate for 100 video frames. Thus, for obtaining HR from a video clip of the desired duration, we obtain an average of the HR estimates for multiple smaller video clips, each of 100 frames. Table 1 shows our results. It indicates that our method outperforms earlier works consisting of 2SR [58], ICA [45] and chrominance-rPPG [10] because they have used BSS for estimating the pulse signal. These techniques are incapable of denoising because they utilize handcrafted representations to model noise and the lack of supervision limits their capability to understand the noise features induced by facial movements [61].

Similarly, our method provides better performance than AHRE [18] and Fusion-EL [19] because they have also used BSS for estimating pulse signals which limit their ability to extract pulse signal attributes. Additionally, they use the same constraints for denoising the temporal signals obtained from all the face ROIs. However, different face regions have local noise sources [3]. This issue is mitigated by the signal embeddings employed in our proposed method since the signal embedding from a particular face region remains unaffected by other facial regions. For the same reasons as above, we obtain better performance than deep learning based approaches Deepphys [9], and HR-CNN [52] due to better representation modeling of physiological sources. Additionally, the aforementioned CNN-based architectures are affected by the noise induced for a short duration due to small facial movements. However, our Transformer-based architecture is able to mitigate such issues. This behavior is attributed to the global processing capabilities of Transformers. Our method outperforms META-rPPG [33] because they use Long Short-Term Memory (LSTM) network for modeling rPPG information, and it is observed that LSTM architectures are prone to information loss for long sequences [13].

The results indicate that our method outperforms

Table 2. Performance evaluation of our proposed method for different number of ROIs. All the values are in BPM

ROIs	UBFC-rPPG			COHFACE		
	σ	MAE	RMSE	σ	MAE	RMSE
8	07.64	05.45	09.39	09.40	11.33	14.72
10	04.99	04.55	06.75	07.67	09.60	12.29
12	03.45	02.91	04.52	07.41	08.01	10.12
14	04.19	03.05	05.19	07.12	10.14	12.39
16	05.00	03.23	05.96	08.04	10.85	13.73

Transformer-based InstTrans architecture [47] because they utilize difference of frames for obtaining the temporal signals and a dual-stream architecture for identifying facial regions with significant rPPG information. Thus, it requires large-scale datasets for overcoming underfitting [7]. Moreover, it has not utilized pre-training for alleviating the effect of poor inductive bias of the Transformer architecture [12]. In contrast, our efficient temporal signal embeddings, pre-training, and data augmentation techniques, followed by the Transformer-based architecture, allows our method to overcome these issues and provide correct HR estimation.

4.4. Ablation Study

This subsection provides the impact of training parameters on our proposed architecture, *RADIANT* and the importance of different components of our architecture. Initially, we used different numbers of ROIs for pulse signal estimation in our proposed method, and the results are reported in Table 2. It can be observed that efficacy first improves when the number of ROIs increases. Increasing the number of ROIs increases the number of temporal signals, allowing better feature representation learning for the physiological signals. Whereas including more facial ROIs will also cause an increase in noise components, which will affect the performance of our architecture. Thus, we obtain optimal results for 12 facial ROIs and observe a decrease in performance when more than 12 ROIs are used.

Table 3. Performance evaluation of the proposed method for different experimental settings. All the values are in BPM and all the metrics represent better performance if they have lower values.

	UBFC-rPPG			COHFACE		
	σ	MAE	RMSE	σ	MAE	RMSE
<i>Ours</i>	03.45	02.91	04.52	07.41	08.01	10.12
<i>GS</i>	13.74	48.50	50.41	11.26	17.76	21.03
<i>RGB</i>	13.76	46.46	48.45	11.34	27.06	29.34
<i>NP</i>	08.13	07.73	11.22	10.83	13.43	17.26
<i>NS</i>	07.40	06.11	07.98	11.01	13.40	18.61
<i>Conv</i>	07.73	06.25	08.74	11.31	13.82	19.65

Table 3 presents the results for understanding the importance of different components of our method. These experiments are formed by modifying the proposed method, *RADIANT*. The experiments *GS* and *RGB* are formed by

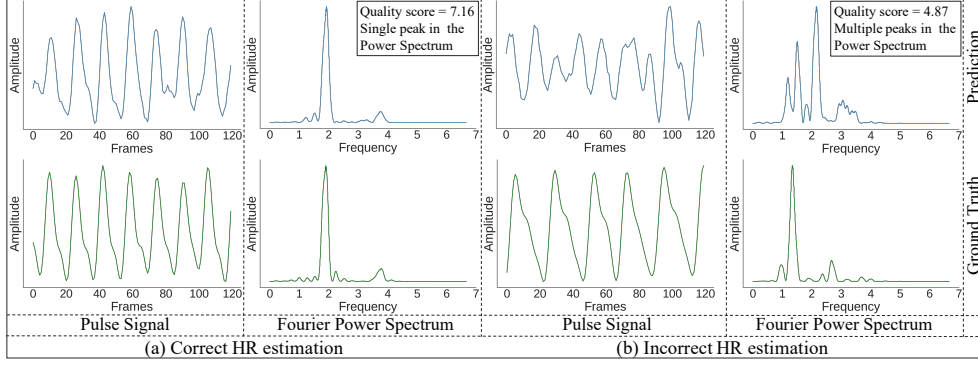


Figure 3. (a) Example of successful HR estimation and (b) example of unsuccessful HR estimation.

replacing the chrominance signals with the green and RGB channel’s temporal signals, respectively. The results show that the chrominance signals outperform the other representations because they suppress the effects of motion and luminance variations [10]. The *NP* experiment is created by replacing the mean squared error loss function with the negative pearson loss function. The results show that our proposed method outperforms *NP* experiment, indicating that the mean squared error loss function performs better than the negative pearson loss function. In the experiment *NS*, we have used our architecture without pre-training. The results demonstrate that we obtain better performance when our Transformer is pre-trained with synthetic temporal signals because synthetic signals provide the necessary domain adaptation capability to our architecture for pulse estimation. Similarly, the experiment *Conv* is formed by replacing the MLP projection layer by the input Convolutional layer of kernel (7×7), stride (2×2) and padding of 3 as used in ResNet-18 [23] input. Utilizing such projection leads to embeddings of d_c dimensions. Thus, we have applied an additional max pooling layer with kernel and stride of ($1 \times d_c$) so that we obtain signal embeddings of the required dimensions by the Transformer layer. The results show that we perform poorly when we utilize information from nearby temporal signals for signal embeddings using the Convolutional layer. It happens because the noise from other temporal signals interferes with the rPPG information.

4.5. Discussion

We depict an example of HR estimation by our proposed method in Figure 3. The figure compares the predicted and ground truth pulse for success and failure cases. The first row represents the estimated pulse and its Fourier power spectrum, and the second row shows the ground truth. For the success case in Figure 3 (a), we observe that the estimated pulse signal correlates well with the ground truth. However, we observe a deviation in the estimated pulse signal from the ground truth signal for certain video frames be-

cause these frames contain slight noise due to facial movements. In contrast, the effect of large facial movements on the estimated pulse signal can be observed in Figure 3 (b). Comparison of the quality scores for the respective temporal signals suggests that it is an important parameter determining the quality of pulse estimation.

5. Conclusion

The existing deep learning based rPPG approaches have suffered from underfitting on limited datasets and failed in modeling long temporal dependencies leading to incorrect HR estimation. To mitigate these issues, we have provided a Transformer-based pulse estimation method, *RA-DIANT*. The architecture has benefitted from the attention processing capabilities of the Transformer that allows modeling of long temporal dependencies and effective denoising for correct pulse estimation. Further, it has utilized an MLP projection for obtaining the signal embeddings that provide an adequate subspace for rPPG feature representation. Our experiments justified the isolation of rPPG information in the signal embeddings. We have investigated the possibility of utilizing pre-training and efficient data augmentation techniques to improve generalization capabilities. The experimental results have demonstrated that our architecture provides better results when pre-trained on the synthetic dataset. Also, it indicates that data augmentation allows our architecture to generalize well over the normal HR range. Our results on extensively utilized datasets have shown that our architecture outperforms previous well-known rPPG methods. In the future, we will be working on providing an end-to-end HR estimation network by automating the temporal signal extraction process.

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